

Student's 't' test for difference of means

To test the significant difference between 2 means $\bar{x}_1$ & $\bar{x}_2$ of sample sizes n_1 and n_2 .

Test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \text{ where } s^2 = \frac{\sum(x_1 - \bar{x}_1)^2 + \sum(x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2} \text{ (Or) } s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

s_1, s_2 = sample standard deviation, **degrees of freedom = $n_1 + n_2 - 2$**

Problems:

- 1. The average numbers of articles produced by two machines per day are 200 & 250 with standard deviation 20 and 25 respectively on the basis of records of 25 days production. Can you regard both the machines equally efficient at 1% level of significance?**

Given

$$n_1 = 25 ; \quad n_2 = 25$$

$$\bar{x}_1 = 200 ; \quad \bar{x}_2 = 250$$

$$s_1 = 20 ; \quad s_2 = 25$$

$$\therefore s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \frac{1}{48} [25(20)^2 + 25(25)^2] = 533.25$$

$$s = 23.10$$

Null Hypothesis H_0 : Both the machines are equally efficient

$$\text{i.e., } \mu_1 = \mu_2$$

Alternative hypothesis: H_1 : $\mu_1 \neq \mu_2$

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$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{200 - 250}{2310 \sqrt{\frac{1}{25} + \frac{1}{25}}} = -7.65$$

$$|t| = 7.65$$

Table value: Degrees of freedom: $n_1 + n_2 - 2 = 48$ d.f at 1% level of significance is 2.58.

Conclusion: Calculated value > tabulated value

We reject H_0 .

i.e., Both machines are not equally efficient at 1% level of significance.

2. The means of 2 random samples of size 9 and 7 are 196.42 and 198.82 respectively. The sum of the squares of the deviation from the mean is 26.94 and 18.73 respectively. Can you regard both the samples have been drawn from the same normal population?

Given

$$n_1 = 9; \quad n_2 = 7$$

$$\bar{x}_1 = 196.42; \quad \bar{x}_2 = 198.82$$

$$\sum (x_1 - \bar{x}_1)^2 = 26.94; \quad \sum (x_2 - \bar{x}_2)^2 = 18.73$$

$$s^2 = \frac{\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2} = \frac{26.94 + 18.73}{9 + 7 - 2} = 3.26$$

$$s = 1.81$$

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Null Hypothesis H_0 : The two samples are drawn from the same population. i.e., $\mu_1 = \mu_2$

Alternative hypothesis: H_1 : $\mu_1 \neq \mu_2$

Test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{19642 - 19882}{1.81 \sqrt{\frac{1}{9} + \frac{1}{7}}} = -2.63$$

$$|t| = 2.63$$

Table value: Degrees of freedom: $n_1 + n_2 - 2 = 14$ d.f at 5% level of significance is 2.15.

Conclusion: Calculated value > tabulated value

We reject H_0 .

The 2 samples are not drawn from the same population.

Practice problem:

- Below are given the gain in weights of pigs fed on 2 diets A and B.

diet A	25	32	30	34	24	14	32	24	30	31	35	25	-	-	-
diet B	44	34	22	10	47	31	40	30	32	35	18	21	35	29	22

Test if the 2 diets differ significantly as regards their effect on increase in weight.

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